CIRCUITS SYSTEMS SIGNAL PROCESS
VOL. 7, NO. 3, 1988

MINIMIZATION OF TRANSIENT SIGNALS IN RECURSIVE TIME-VARYING DIGITAL FILTERS*

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Abstract. Changing the coefficient set of a recursive digital filter during operation results in a transient signal at the filter output. The amplitude of this transient signal may exceed the amplitude of the output signal before and well after the change of coefficients. This effect is rather disturbing in some applications, such as the processing of audio signals. It is shown how these transients can be minimized without increasing the computational complexity of the filter by the use of a properly chosen set of intermediate coefficients.

1. Introduction

We consider recursive digital filters with coefficients which can be changed during operation. Each change of the coefficient set causes a transient signal at the filter output. Depending on the coefficient sets and the input signal, this transient signal may attain values which exceed the amplitude of the output signal before and well after a coefficient change. Transient signals of this kind are highly undesirable, since they may be mistaken as parts of the input signal. For instance, when processing audio signals, they are audible as click-like noises. A practical aspect of this problem is the design of digital mixing desks (see [8]). They require adjustable (i.e., time-varying) filters for the on-line modification of the sound spectrum. Of course, switching noises cannot be tolerated.

An example is shown in Figure 1. Figure 1(a) and (b) shows the steady-state output signals of two constant coefficient second-order filters with input signal $v(k) = \sin(\pi/4)k$ and two different coefficient sets. Figure 1(c) shows the output of a second-order filter if its coefficients are changed at $k = M$ from set (a) to set (b). Obviously the values of the transient signal

* Received June 1, 1987.

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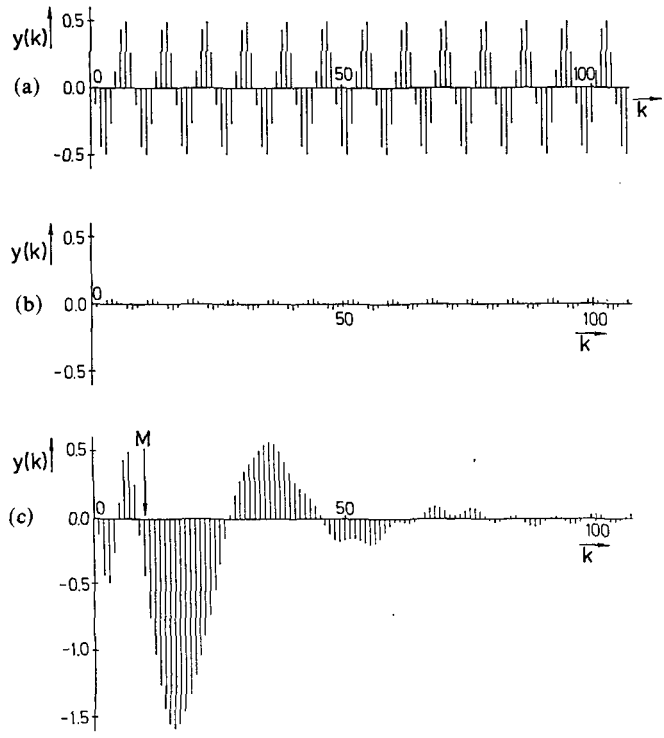


Figure 1. Response to an input signal $v(k) = \sin(\pi/4)k$. (a) Start coefficient set, (b) end coefficient set, (c) direct change of coefficients at $k = M$.

far exceed the amplitudes of the steady-state signals of Figure 1(a) and (b). (The coefficient sets are given in Appendix 1.)

In this paper we analyze the situation and propose ways to reduce transient signals. It is assumed that the spacing between coefficient changes is large enough such that subsequent transient signals do not interfere. Thus it is sufficient to consider only a single coefficient change. Time-varying recursive filters with interfering coefficient changes have been treated in [6].

The considerations which follow apply to recursive filters in state-space realization. In Section 2 we compile some useful relationships for their description. They are used in Section 3 to analyze the time-varying filters under consideration, either with a direct change of coefficients or with an intermediate coefficient set. These intermediate coefficients are derived to fulfill certain minimization criteria for the transient signals in Section 4. Examples are presented in Section 5.

2. Time-invariant recursive filters

Consider a single-input, single-output filter with minimal $(\mathbf{A}, \mathbf{b}, \mathbf{c}, d)$

realization (e.g., [1])

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{b}v(k), \tag{1a}$$

$$y(k) = \mathbf{c}^{\mathrm{T}}\mathbf{x}(k) + dv(k). \tag{1b}$$

$v(k)$ and $y(k)$ are the input and output sequences, $\mathbf{x}(k)$ is the n-dimensional state vector, and $\mathbf{A}$, $\mathbf{b}$, $\mathbf{c}^{\mathrm{T}}$, and d are matrices and vectors of appropriate size (bold face capital letters denote matrices, bold face lowercase letters denote vectors, and the superscript T denotes transposition). They can be composed to a block matrix

$$\mathbf{S} = \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{c}^{\mathrm{T}} & d \end{bmatrix}. \tag{2}$$

Equations (1a) and (1b) can now be combined to give

$$\begin{bmatrix} \mathbf{x}(k+1) \\ y(k) \end{bmatrix} = \mathbf{S} \cdot \begin{bmatrix} \mathbf{x}(k) \\ v(k) \end{bmatrix}. \tag{1c}$$

We introduce the sequences

$$\mathbf{f}(k) = \begin{cases} 0, & k \le 0, \\ \mathbf{A}^{k-1}\mathbf{b}, & k > 0, \end{cases} \tag{3a}$$

$$\mathbf{g}^{\mathrm{T}}(k) = \begin{cases} 0, & k \le 0, \\ \mathbf{c}^{\mathrm{T}}\mathbf{A}^{k-1}, & k > 0, \end{cases} \tag{3b}$$

$$h_0(k) = \begin{cases} 0, & k < 0, \\ d, & k = 0, \\ \mathbf{c}^{\mathrm{T}}\mathbf{A}^{k-1}\mathbf{b} = \mathbf{c}^{\mathrm{T}}\mathbf{f}(k) = \mathbf{g}^{\mathrm{T}}(k)\mathbf{b}, & k > 0. \end{cases} \tag{3c}$$

$h_0(k)$ is the impulse response from the input to the output, $\mathbf{f}(k)$ is the vector of impulse responses from the input to the state variables, and $\mathbf{g}^{\mathrm{T}}(k)$ describes the reaction of the system to an initial state and zero input [2].

The solution of the state equations (1) to an initial state $\mathbf{x}(k_0)$ at $k = k_0$ and an input signal $v(k)$ is then given by (see [1])

$$\mathbf{x}(k) = \mathbf{A}^{k-k_0}\mathbf{x}(k_0) + \sum_{\kappa=k_0}^{k} \mathbf{f}(k-\kappa)v(\kappa), \tag{4a}$$

$$y(k) = \mathbf{g}^{\mathrm{T}}(k-k_0+1)\mathbf{x}(k_0) + \sum_{\kappa=k_0}^{k} h_0(k-\kappa)v(\kappa). \tag{4b}$$

We obtain, for $k_0 \to -\infty$,

$$\mathbf{x}(k) = \sum_{\kappa=-\infty}^{k} \mathbf{f}(k-\kappa)v(\kappa) = \mathbf{f}(k) * v(k), \tag{5a}$$

$$y(k) = \sum_{\kappa=-\infty}^{k} h_0(k-\kappa)v(\kappa) = h_0(k) * v(k). \tag{5b}$$

3. Time-varying recursive filters

In this section we consider in detail a recursive filter with coefficients that can be changed during operation and compare its output signal to that of the corresponding constant coefficient filters. Figure 2 shows an arrangement of these filters and their output signals.

$$\mathbf{S}_- = \begin{bmatrix} \mathbf{A}_- & \mathbf{b}_- \\ \mathbf{c}_-^T & d_- \end{bmatrix} \tag{6a}$$

and

$$\mathbf{S}_+ = \begin{bmatrix} \mathbf{A}_+ & \mathbf{b}_+ \\ \mathbf{c}_+^T & d_+ \end{bmatrix} \tag{6b}$$

denote the coefficient sets of the start and end filters, respectively. $y_-(k)$ and $y_+(k)$ are their individual output signals, if they operate separately.

$\mathbf{S}_{-/+}$ denotes a time-varying recursive filter, the coefficient set of which is changed from $\mathbf{S}_-$ to $\mathbf{S}_+$ at $k=0$. $y(k)$ is its output signal. The indices $-$ and $+$ show that the corresponding coefficient sets are valid for negative and positive k, respectively. As an example, let $\mathbf{S}_-$ and $\mathbf{S}_+$ be chosen as in Appendix 1. The resulting signals $y_-(k)$, $y_+(k)$, and $y(k)$ are shown in figure 1(a)-(c), respectively.

We now introduce the difference signals

$$\Delta_-(k) = y(k) - y_-(k) \qquad (=0,\ k<0) \tag{7a}$$

and

$$\Delta_+(k) = y(k) - y_+(k) \qquad (=0,\ k\to\infty) \tag{7b}$$

as a description for the transient signal.

In order to obtain these difference signals from the start and end coefficient sets, the state equations (1) have to be generalized such that they describe time-varying systems. They are given by (see [1])

$$\begin{bmatrix} \mathbf{x}(k+1) \\ y(k) \end{bmatrix} = \mathbf{S}(k) \begin{bmatrix} \mathbf{x}(k) \\ v(k) \end{bmatrix} \tag{8}$$

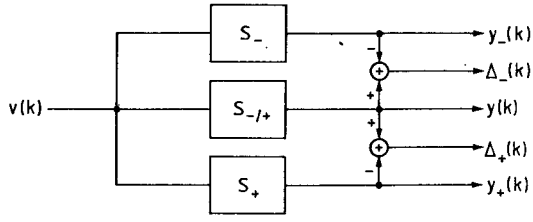


Figure 2. Time-varying filter $\mathbf{S}_{-/+}$ compared with the time-invariant filters $\mathbf{S}_-$ and $\mathbf{S}_+$.

with

$$\mathbf{S}(k)=\begin{bmatrix}\mathbf{A}(k) & \mathbf{b}(k)\\ \mathbf{c}^{\mathrm{T}}(k) & d(k)\end{bmatrix}. \tag{9}$$

3.1. *Direct coefficient change*

First we consider the case when the coefficients of the time-varying filter are changed in one filter step, that is

$$\mathbf{S}(k)=\begin{cases}\mathbf{S}_-, & k<0,\\ \mathbf{S}_+, & k\geq 0.\end{cases} \tag{10}$$

Since the coefficients of the time-varying filter are piecewise constant, we can use the relations of Section 2 to describe the output signals of the filters in Figure 2. The subscripts $-$ and $+$ used in (6) also apply to the functions defined in (3).

The output signals $y_-(k)$ and $y_+(k)$ could be written according to (5b), but we will use (5a) for $k=0$:

$$\mathbf{x}_-(0)=\sum_{\kappa=-\infty}^{0}\mathbf{f}_-(-\kappa)v(\kappa), \tag{11a}$$

$$\mathbf{x}_+(0)=\sum_{\kappa=-\infty}^{0}\mathbf{f}_+(-\kappa)v(\kappa) \tag{11b}$$

and insert into (4b) for $k_0=0$:

$$y_-(k)=\mathbf{g}_-^{\mathrm{T}}(k+1)\mathbf{x}_-(0)+\sum_{\kappa=0}^{k}h_{0-}(k-\kappa)v(\kappa), \tag{12a}$$

$$y_+(k)=\mathbf{g}_+^{\mathrm{T}}(k+1)\mathbf{x}_+(0)+\sum_{\kappa=0}^{k}h_{0+}(k-\kappa)v(\kappa). \tag{12b}$$

We proceed in the same way for the time-varying filter. It is identical to $\mathbf{S}_-$ for $k<0$ and thus

$$\mathbf{x}(0)=\mathbf{x}_-(0). \tag{11c}$$

For $k\geq 0$ it is identical to $\mathbf{S}_+$ and therefore, from (4b),

$$y(k)=\mathbf{g}_+^{\mathrm{T}}(k+1)\mathbf{x}_-(0)+\sum_{\kappa=0}^{k}h_{0+}(k-\kappa)v(\kappa). \tag{12c}$$

For the difference signals it follows for $k\geq 0$ that

$$\Delta_-(k)=[\mathbf{g}_+^{\mathrm{T}}(k+1)-\mathbf{g}_-^{\mathrm{T}}(k+1)]\mathbf{x}_-(0)+\sum_{\kappa=0}^{k}[h_{0+}(k-\kappa)-h_{0-}(k-\kappa)]\cdot v(\kappa) \tag{13a}$$

and

$$\Delta_+(k) = \mathbf{g}_+^T(k+1)[\mathbf{x}_-(0) - \mathbf{x}_+(0)]. \tag{13b}$$

3.2. *Use of an intermediate coefficient set*

With a direct change of coefficients as in Section 3.1, the difference signals depend only on the start and the end coefficient set. In order to gain degrees of freedom for the minimization of the difference signals, we introduce an intermediate coefficient set,

$$\mathbf{S}_0 = \begin{bmatrix} \mathbf{A}_0 & \mathbf{b}_0 \\ \mathbf{c}_0^T & d_0 \end{bmatrix}, \tag{14}$$

which is used for $k = 0$, such that the coefficients of the time-varying filter are now given by

$$\mathbf{S}(k) = \begin{cases} \mathbf{S}_- & \text{for } k < 0, \\ \mathbf{S}_0 & \text{for } k = 0, \\ \mathbf{S}_+ & \text{for } k > 0. \end{cases} \tag{15}$$

Its output signal for $k < 0$ is given by

$$y(k) = y_-(k) = \sum_{\kappa=-\infty}^{k} h_{0-}(k-\kappa)v(\kappa), \qquad k < 0, \tag{16a}$$

and again

$$\mathbf{x}(0) = \mathbf{x}_-(0) \tag{17a}$$

The state equations yield for $k = 0$

$$\mathbf{x}(1) = \mathbf{A}_0\mathbf{x}_-(0) + \mathbf{b}_0 v(0), \tag{17b}$$

$$y(0) = \mathbf{c}_0^T\mathbf{x}_-(0) + d_0 v(0). \tag{16b}$$

For $k > 0$ it follows from (4b) with $k_0 = 1$ and (17b) that

$$y(k) = \mathbf{g}_+^T(k)\mathbf{A}_0\mathbf{x}_-(0) + \mathbf{g}_+^T(k)\mathbf{b}_0 v(0) + \sum_{\kappa=1}^{k} h_{0+}(k-\kappa)v(\kappa). \tag{16c}$$

By comparing (16) and (12) the difference signals turn out to be

$$\Delta_-(k) = \begin{cases} 0, & k < 0, \\ [\mathbf{c}_0^T - \mathbf{c}_-^T]\mathbf{x}_-(0) + [d_0 - d_-]v(0), & k = 0, \\ [\mathbf{g}_+^T(k)\mathbf{A}_0 - \mathbf{g}_-^T(k+1)]\mathbf{x}_-(0) + [\mathbf{g}_+^T(k)\mathbf{b}_0 - h_{0-}(k)]v(0) & \\ \quad + \sum_{\kappa=1}^{k} [h_{0+}(k-\kappa) - h_{0-}(k-\kappa)]v(k), & k > 0, \end{cases} \tag{18a}$$

$$\Delta_+(k)=\begin{cases}\sum_{\kappa=-\infty}^{k}[h_{0-}(k-\kappa)-h_{0+}(k-\kappa)]v(\kappa), & k<0,\\ [\mathbf{c}_0^{\mathrm{T}}\mathbf{x}_-(0)-\mathbf{c}_+^{\mathrm{T}}\mathbf{x}_+(0)]+[d_0-d_+]v(0), & k=0,\\ \mathbf{g}_+^{\mathrm{T}}(k)[\mathbf{A}_0\mathbf{x}_-(0)-\mathbf{A}_+\mathbf{x}_+(0)]+\mathbf{g}_+^{\mathrm{T}}(k)[\mathbf{b}_0-\mathbf{b}_+]v(0), & k>0.\end{cases} \tag{18b}$$

4. Derivation of the optimal intermediate coefficients

The intermediate coefficient set $\mathbf{S}_0$ introduced in the previous section will now be chosen such that the difference signals become minimal in a certain sense. Of course they do not only depend on the filter coefficients but also on the input signal. We assume that the input is white noise with zero mean and the autocorrelation sequence

$$\varphi_{vv}(\lambda)=E\{v(k)v(k+\lambda)\}=\sigma_v^2\,\gamma_0(\lambda), \tag{19}$$

where

$$\gamma_0(\lambda)=\begin{cases}1, & \lambda=0,\\ 0, & \lambda\neq 0.\end{cases}$$

Then we calculate the variances of the difference signals and determine the intermediate coefficients such that these variances become minimal.

We start with $\Delta_-(0)$. From (18a) and (19) it follows for its variance

$$\begin{aligned}\sigma_{\Delta-}^2(0)_- &= E\{\Delta_-^2(0)\}\\ &=[\mathbf{c}_0-\mathbf{c}_-]^{\mathrm{T}}E\{\mathbf{x}_-(0)\mathbf{x}_-^{\mathrm{T}}(0)\}[\mathbf{x}_0-\mathbf{c}_-]+[d_0-d_-]^2\sigma_v^2.\end{aligned}$$

Equation (12a) yields

$$E\{\mathbf{x}_-(0)\mathbf{x}_-^{\mathrm{T}}(0)\}=\sigma_v^2\mathbf{K}_{--}, \tag{20}$$

where

$$\mathbf{K}_{--}=\sum_{k=0}^{\infty}\mathbf{f}_-(k)\mathbf{f}_-^{\mathrm{T}}(k) \tag{21a}$$

is the covariance-matrix of the system $\mathbf{S}_-$ (e.g., [3]). Thus we have

$$\sigma_\Delta^2(0)/\sigma_v^2=[\mathbf{c}_0-\mathbf{c}_-]^{\mathrm{T}}\mathbf{K}_{--}[\mathbf{c}_0-\mathbf{c}_-]+[d_0-d_-]^2. \tag{22a}$$

We obtain, in the same fashion, for $k>0$

$$\begin{aligned}\sigma_{\Delta-}^2(k)/\sigma_v^2 &=[\mathbf{g}_+^{\mathrm{T}}(k)\mathbf{A}_0-\mathbf{g}_-^{\mathrm{T}}(k+1)]\mathbf{K}_{--}[\mathbf{g}_+^{\mathrm{T}}(k)\mathbf{A}_0-\mathbf{g}_-^{\mathrm{T}}(k+1)]^{\mathrm{T}}\\ &\quad+[\mathbf{g}_+^{\mathrm{T}}(k)\mathbf{b}_0-h_{0-}(k)]^2+c_1(k)\\ &=\mathbf{g}_+^{\mathrm{T}}(k)\mathbf{A}_0\mathbf{K}_{--}\mathbf{A}_0^{\mathrm{T}}\mathbf{g}_+(k)-\mathbf{g}_+^{\mathrm{T}}(k)\mathbf{A}_0\mathbf{K}_{--}\mathbf{g}_-^{\mathrm{T}}(k+1)\\ &\quad-\mathbf{g}_-^{\mathrm{T}}(k+1)\mathbf{K}_{--}\mathbf{A}_0^{\mathrm{T}}\mathbf{g}_+(k)\\ &\quad+\mathbf{g}_+^{\mathrm{T}}(k)\mathbf{b}_0\mathbf{b}_0^{\mathrm{T}}\mathbf{g}_+(k)-2\mathbf{g}_+^{\mathrm{T}}(k)\mathbf{b}_0h_{0-}(k)+c_2(k).\end{aligned} \tag{22b}$$

The terms

$$c_1(k) = \sum_{\kappa=0}^{k-1} [h_{0_+}(\kappa) - h_{0_-}(\kappa)]^2$$

and

$$c_2(k) = \mathbf{g}_-^{\mathrm{T}}(k+1)\mathbf{K}_{--}\mathbf{g}_-^{\mathrm{T}}(k+1) + h_{0_-}^2(k) + c_1(k)$$

do not depend on the intermediate coefficients.

In the same way it follows for the variance of $\Delta_+(k)$

$$\sigma_{\Delta_+}^2(0)/\sigma_v^2 = \mathbf{c}_0^{\mathrm{T}}\mathbf{K}_{--}\mathbf{c}_0 - \mathbf{c}_0^{\mathrm{T}}\mathbf{K}_{-+}\mathbf{c}_+ - \mathbf{c}_+^{\mathrm{T}}\mathbf{K}_{+-}\mathbf{c}_0 + (d_0 - d_+)^2 = c_3(0) \quad (23a)$$

and

$$\begin{aligned}\sigma_{\Delta_+}^2(k)/\sigma_v^2 = {} & \mathbf{g}_+^{\mathrm{T}}(k)\mathbf{A}_0\mathbf{K}_{--}\mathbf{A}_0^{\mathrm{T}}\mathbf{g}_+(k) - \mathbf{g}_+^{\mathrm{T}}(k)\mathbf{A}_0\mathbf{K}_{-+}\mathbf{g}_+(k+1) \\ & - \mathbf{g}_+^{\mathrm{T}}(k+1)\mathbf{K}_{+-}\mathbf{A}_0^{\mathrm{T}}\mathbf{g}_+(k) \\ & + g_+^{T}(k)(\mathbf{b}_0 - \mathbf{b}_+)(\mathbf{b}_0 - \mathbf{b}_+)^{\mathrm{T}}\mathbf{g}_+(k) + c_3(k) \end{aligned} \quad (23b)$$

with the cross-covariance matrices

$$\begin{aligned}\mathbf{K}_{-+} &= \sum_{\kappa=0}^{\infty} \mathbf{f}_-(k)\mathbf{f}_+^{\mathrm{T}}(k), \\ \mathbf{K}_{+-} &= \mathbf{K}_{-+}^{\mathrm{T}}, \end{aligned} \quad (21b)$$

and

$$c_3(k) = \mathbf{g}_+^{\mathrm{T}}(k+1)\mathbf{K}_{++}\mathbf{g}_+(k+1).$$

The next step is to determine the intermediate coefficients such that the variances of the difference signals become minimal in a certain sense. We will consider three different minimization criteria.

4.1. *Minimization of the first values of the variance of* $\Delta_-(k)$

Here we try to determine the intermediate coefficients such that $\sigma_{\Delta_-}^2(k)$ becomes minimal for as many values of k as possible. For $k=0$ it follows immediately from (22a) that

$$\sigma_{\Delta_-}^2(0) = 0 \qquad \text{for} \quad \mathbf{c}_0 = \mathbf{c}_- \quad \text{and} \quad d_0 = d_-.$$

From (22b) it follows that $\sigma_{\Delta_-}^2(1)$ becomes minimal for

$$\mathbf{c}_+^{\mathrm{T}}\mathbf{A}_0 = \mathbf{c}_-^{\mathrm{T}}\mathbf{A}_- \quad \text{and} \quad \mathbf{c}_+^{\mathrm{T}}\mathbf{b}_0 = \mathbf{c}_-^{\mathrm{T}}\mathbf{b}_-$$

and $\sigma^2_{\Delta_-}(2)$ for

$$\mathbf{c}_+^T \mathbf{A}_+ \mathbf{A}_0 = \mathbf{c}_-^T \mathbf{A}_-^2 \quad \text{and} \quad \mathbf{c}_+^T \mathbf{A}_+ \mathbf{b}_0 = \mathbf{c}_-^T \mathbf{A}_- \mathbf{b}_-.$$

The intermediate coefficients turn out to be

$$\mathbf{A}_0 = \begin{bmatrix} \mathbf{c}_+^T \\ \mathbf{c}_+^T \mathbf{A}_+ \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{c}_-^T \\ \mathbf{c}_-^T \mathbf{A}_- \end{bmatrix} \mathbf{A}_-, \tag{24a}$$

$$\mathbf{b}_0 = \begin{bmatrix} \mathbf{c}_+^T \\ \mathbf{c}_+^T \mathbf{A}_+ \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{c}_-^T \\ \mathbf{c}_-^T \mathbf{A}_- \end{bmatrix} \mathbf{b}_-, \tag{24b}$$

$$\mathbf{c}_0 = \mathbf{c}_-, \qquad d_0 = d_-. \tag{24c,d}$$

4.2. *Minimization of the variance of* $\Delta_-(k)$ *for all values*

The intermediate coefficients are determined such that the sum over all their contributions to $\sigma^2_{\Delta_-}(k)$ for $k = 0, 1, \ldots$ becomes minimal. Since $\sigma^2_{\Delta_-}(0) = 0$ for $\mathbf{c}_0 = \mathbf{c}_-$ and $d_0 = d_-$ it suffices to consider

$$\alpha(\mathbf{A}_0, \mathbf{b}_0) = \sum_{k=1}^{\infty} [\sigma^2_{\Delta_-}(k)/\sigma_v^2 - c_2(k)] \quad \text{from (22b).}$$

We derivate this expression with respect to the matrix $\mathbf{A}_0$ and the vector $\mathbf{b}_0$ in order to find the minimal value. The derivation rules for the matrices and vectors used here are compiled in Appendix 2.

We obtain

$$\frac{\partial}{\partial \mathbf{A}_0} \alpha(\mathbf{A}_0, \mathbf{b}_0) = 2[\mathbf{W}_{++} \mathbf{A}_0 - \mathbf{W}_{+-} \mathbf{A}_-] \mathbf{K}_{--},$$

$$\frac{\partial}{\partial \mathbf{b}_0} \alpha(\mathbf{A}_0, \mathbf{b}_0) = 2[\mathbf{W}_{++} \mathbf{b}_0 - \mathbf{W}_{+-} \mathbf{b}_-]$$

with the (cross-) covariance matrices

$$\mathbf{W}_{++} = \sum_{k=0}^{\infty} \mathbf{g}_+(k) \mathbf{g}_+^T(k)$$

and $\mathbf{W}_{+-}$ alike (compare (21a, b)).

Setting the derivatives to zero gives the intermediate coefficients for this case:

$$\mathbf{A}_0 = \mathbf{W}_{++}^{-1} \mathbf{W}_{+-} \mathbf{A}_-, \tag{25a}$$

$$\mathbf{b}_0 = \mathbf{W}_{++}^{-1} \mathbf{W}_{+-} \mathbf{b}_-, \tag{25b}$$

$$\mathbf{c}_0 = \mathbf{c}_-, \tag{25c}$$

$$d_0 = d_-. \tag{25d}$$

4.3. *Minimization of the variance of* $\Delta_+(k)$

In this case the minimization of $\Delta_+(k)$ value by value or for all values are identical. Equation (23) gives immediately $\mathbf{b}_0 = \mathbf{b}_+$ and $d_0 = d_+$. $\mathbf{A}_0$ and $\mathbf{c}_0$

follow again from the derivation:

$$\frac{\partial}{\partial \mathbf{c}_0} \sigma^2_{\Delta_+}(0)/\sigma^2_v = 2(\mathbf{K}_{--}\mathbf{c}_0 - \mathbf{K}_{-+}\mathbf{c}_+),$$

$$\frac{\partial}{\partial \mathbf{A}_0} \sigma^2_{\Delta_+}(k)/\sigma^2_v = 2\mathbf{g}_+(k)\mathbf{g}^{\mathrm{T}}_+(k)(\mathbf{A}_0\mathbf{K}_{--} - \mathbf{A}_+\mathbf{K}_{+-}).$$

The intermediate coefficients are

$$\mathbf{A}_0 = \mathbf{A}_+\mathbf{K}_{+-}\mathbf{K}_{--}^{-1}, \tag{26a}$$

$$\mathbf{b}_0 = \mathbf{b}_+, \tag{26b}$$

$$\mathbf{c}_0^{\mathrm{T}} = \mathbf{c}_+^{\mathrm{T}}\mathbf{K}_{+-}\mathbf{K}_{--}^{-1}, \tag{26c}$$

$$d_0 = d_+. \tag{26d}$$

4.4. *Compilation of the results*

We compile the results for the three minimization criteria with the help of (2) in a unifying form.

Minimization of the variance of $\Delta_-(k)$:

$$\mathbf{S}_0 = \begin{bmatrix} \mathbf{T} & 0 \\ 0 & 1 \end{bmatrix} \mathbf{S}_-,$$

where

$$\mathbf{T} = \begin{bmatrix} \mathbf{c}_+^{\mathrm{T}} \\ \mathbf{c}_+^{\mathrm{T}}\mathbf{A}_+ \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{c}_-^{\mathrm{T}} \\ \mathbf{c}_-^{\mathrm{T}}\mathbf{A}_- \end{bmatrix}$$

for the minimization of the first values and

$$\mathbf{T} = \mathbf{W}_{++}^{-1}\mathbf{W}_{+-}$$

for the minimization for all values.

Minimization of the variance of $d_+(k)$:

$$\mathbf{S}_0 = \mathbf{S}_+ \begin{bmatrix} \mathbf{T} & 0 \\ 0 & 1 \end{bmatrix}$$

with

$$\mathbf{T} = \mathbf{K}_{+-}\mathbf{K}_{--}^{-1}.$$

5. Examples

As an example we consider the coefficients $\mathbf{S}_-$ and $\mathbf{S}_+$ given in Appendix 1. At first the input is assumed to be a white noise signal with variance $\sigma_v^2 = \frac{1}{3}$. Figure 3 shows the variances of $\Delta_-(k)$ and $\Delta_+(k)$ for the cases

(a) direct coefficient change

and use of a set of intermediate coefficients according to minimization of

(b) the first values of $\sigma^2_{\Delta_-}(k)$,

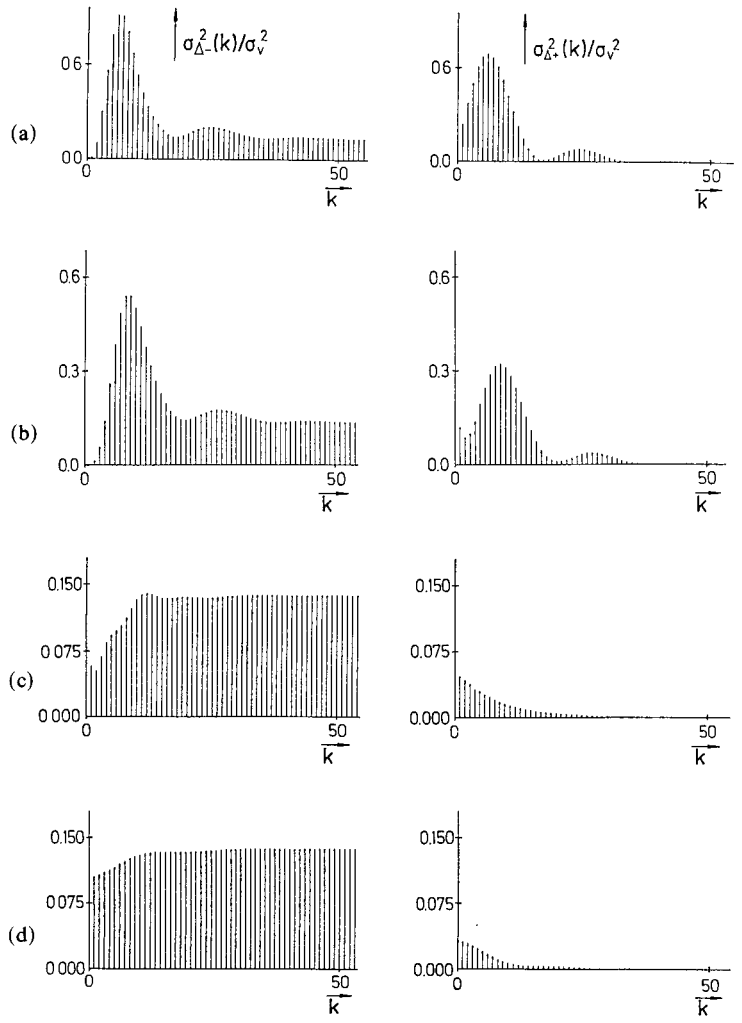


Figure 3. Response to a white noise signal. (a) Direct coefficient change. Intermediate coefficients: (b) minimization of the first values of $\sigma^2_{\Delta_-}(k)$, (c) minimization of $\sigma^2_{\Delta_-}(k)$ for all values, (d) minimization of $\sigma^2_{\Delta_+}(k)$.

(c) $\sigma^2_{\Delta_-}(k)$ for all values, and
(d) $\sigma^2_{+}(k)$.

Obviously, the direct coefficient change produces a large transient signal. It is reduced by the intermediate coefficients according to (b), but not significantly. However, note that the first values of $\sigma^2_{\Delta_-}(k)$ are indeed very small. The choice of the intermediate coefficients according to (c) or (d) avoids any increase of the variance during the transition period.

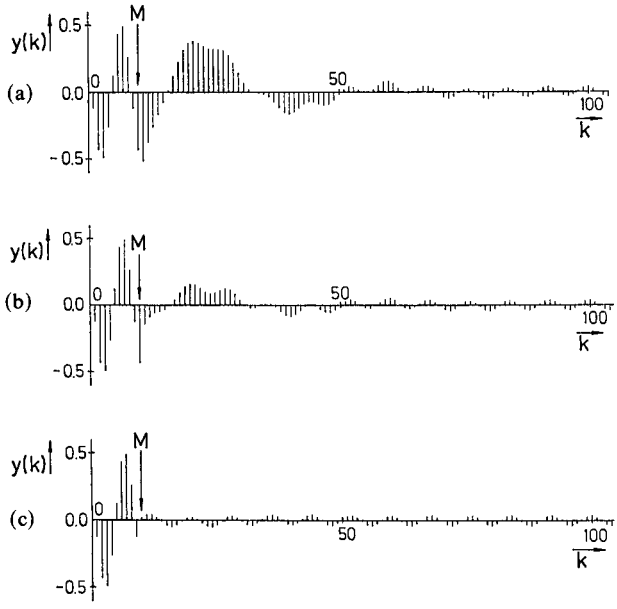


Figure 4. Response to an input signal $v(k) = \sin(\pi/4)k$ with the use of intermediate coefficient sets. (a) Minimization of the first values of $\sigma^2_{\Delta_-}(k)$, (b) minimization of $\sigma^2_{\Delta_-}(k)$ for all values, (c) minimization of $\sigma^2_{\Delta_+}(k)$.

While the use of white noise as the input signal is a good model for broad-band signals, we still have to test the performance of the proposed intermediate coefficients for narrow-band signals. We have already seen, in Figure 1, that the direct change from $\mathbf{S}_-$ to $\mathbf{S}_+$ causes large transients for a sinusoidal input. Figure 4 shows the output signal of the time-varying filter for the same input $v(k) = \sin(\pi/4)k$ but with the use of intermediate coefficients. The minimization of the first values of $\sigma^2_{\Delta_-}(k)$ (Figure 4(a)) reduces the transients effectively but still not satisfactory. Minimization of $\sigma^2_{\Delta_-}(k)$ for all values (Figure 4(b)) does a better job, but the best results are obtained with the minimization of $\sigma^2_{\Delta_+}(k)$ (Figure 4(c)). No transients are visible in this scale.

6. Conclusion

A method has been proposed that reduces transient signals occurring at the output of time-varying recursive filters. It works for narrow-band as well as for broad-band signals. Other than the examples shown here, the effectiveness of this method has also been demonstrated for other sets of start and end coefficients and for different filter structures. Its practical use has also been checked in listening tests performed on high-quality audio signals taken from a compact disc.

The method requires no extra arithmetic operations for the change of coefficients. The reduction of the transient signals is achieved only by the use of one carefully chosen intermediate coefficient set.

Three different minimization criteria have been given, where the minimization of the variance of the difference signal $\Delta_+(k)$ has proven to be the most successful. Two of the three criteria require the calculation of (cross-) covariance matrices. Some methods are given in Appendix 3.

A patent for the method has been applied for [9].

Acknowledgment

A number of people have contributed to the results reported here. Professor Schüßler initiated this research through various valuable suggestions. Dipl.-Ing. H. Göckler drew the author's attention to the practical side of this problem and organized listening tests. Dipl.-Ing. B. Lang did the programming and provided the graphic results in his student thesis [7]. The author wishes to thank all of them.

Appendix 1

The examples in Figures 1, 3, and 4 were obtained with the following coefficients:

start coefficient set	end coefficient set
$\mathbf{A}_- = \begin{bmatrix} 1.32208 & 1 \\ -0.67285 & 0 \end{bmatrix}$,	$\mathbf{A}_+ = \begin{bmatrix} 1.85870 & 1 \\ -0.89025 & 0 \end{bmatrix}$,
$\mathbf{b}_- = \begin{bmatrix} 0.12841363 \\ 0.03177608 \end{bmatrix}$,	$\mathbf{b}_+ = \begin{bmatrix} 0.00824431 \\ 0.00715021 \end{bmatrix}$,
$\mathbf{c}_- = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$,	$\mathbf{c}_+ = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$,
$d_- = 0.09713.$	$d_+ = 0.06515.$

Appendix 2

In Sections 4.2 and 4.3 use has been made of the following rules for the derivation with respect to a vector or to a matrix [4]:

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{A}\mathbf{x} = \mathbf{A}^{\mathrm{T}}, \qquad \frac{\partial}{\partial \mathbf{x}} \mathbf{x}^{\mathrm{T}}\mathbf{A} = \mathbf{A},$$

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{x}^{\mathrm{T}}\mathbf{A}\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{A}^{\mathrm{T}}\mathbf{x},$$

$$\frac{\partial}{\partial \mathbf{A}} \mathbf{y}^{\mathrm{T}} \mathbf{A} \mathbf{v} = \mathbf{y}\mathbf{v}^{\mathrm{T}}, \qquad \frac{\partial}{\partial \mathbf{A}} \mathbf{y}^{\mathrm{T}} \mathbf{A}^{\mathrm{T}} \mathbf{v} = \mathbf{v}\mathbf{y}^{\mathrm{T}},$$

$$\frac{\partial}{\partial \mathbf{A}} \mathbf{y}^{\mathrm{T}} \mathbf{A} \mathbf{B} \mathbf{A}^{\mathrm{T}} \mathbf{v} = \mathbf{y}\mathbf{v}^{\mathrm{T}} \mathbf{A}\mathbf{B}^{\mathrm{T}} + \mathbf{v}\mathbf{y}^{\mathrm{T}} \mathbf{A}\mathbf{B}.$$

Appendix 3

Calculation of the (cross-) covariance matrices

A review of the calculation methods of covariance matrices such as $\mathbf{K}_{--}$ and $\mathbf{W}_{++}$ has been given in [2]. For calculation of the cross-covariance matrices, we have used the following simple algorithm:

$\mathbf{K}_{+-}$ can be shown to satisfy

$$\mathbf{K}_{+-} = \mathbf{A}_{+} \mathbf{K}_{+-} \mathbf{A}_{-}^{\mathrm{T}} + \mathbf{b}_{+} \mathbf{b}_{-}^{\mathrm{T}}.$$

The use of the column-stacking operator

$$\operatorname{vek} \mathbf{K}_{+-} = \begin{bmatrix} \mathbf{k}_1 \\ \vdots \\ \mathbf{k}_n \end{bmatrix},$$

where $\mathbf{k}_1, \ldots, \mathbf{k}_n$ are the column-vectors of $\mathbf{K}_{+-}$, together with the rules for the Kronecker-product $\otimes$ (see [4]) yields the following system of linear equations:

$$[\mathbf{E} - (\mathbf{A}_{+} \otimes \mathbf{A}_{-})] \operatorname{vek} \mathbf{K}_{+-} = \operatorname{vek}(\mathbf{b}_{+} \mathbf{b}_{-}^{\mathrm{T}}).$$

$\mathbf{W}_{+-}$ may be computed in the same way by replacing $\mathbf{A}_{\pm}$ by $\mathbf{A}_{\pm}^{\mathrm{T}}$ and $\mathbf{b}_{\pm}$ by $\mathbf{c}_{\pm}$.

A closed form solution is possible if the matrices $\mathbf{A}_{\pm}$ are normal. It is based on spectral decomposition [5] of the A-matrix and is shown here for second-order systems with real and conjugate complex poles. The elements of $\mathbf{K}_{+-}$ are

$$\mathbf{K}_{+-} = \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix}.$$

Real poles. The general form of the start and the end coefficient sets is

$$\mathbf{A}_{-} = \begin{bmatrix} a_1^{-} & 0 \\ 0 & a_2^{-} \end{bmatrix}, \quad \mathbf{b}_{-} = \begin{bmatrix} b_1^{-} \\ b_2^{-} \end{bmatrix}, \quad \mathbf{c}_{-}^{\mathrm{T}} = [c_1^{-} \quad c_2^{-}], \quad d^{-},$$

$$\mathbf{A}_{+} = \begin{bmatrix} a_1^{+} & 0 \\ 0 & a_2^{+} \end{bmatrix}, \quad \mathbf{b}_{+} = \begin{bmatrix} b_1^{+} \\ b_2^{+} \end{bmatrix}, \quad \mathbf{c}_{+}^{\mathrm{T}} = [c_1^{+} \quad c_2^{+}], \quad d^{+}.$$

The elements of $\mathbf{K}_{+-}$ are given by

$$k_{ij} = \frac{b_i^{+} b_j^{-}}{1 - a_i^{+} a_j^{-}}, \qquad i, j = 1, 2.$$

Conjugate complex poles. The general form of the start and the end coefficient sets is

$$\mathbf{A}_- = \begin{bmatrix} a_1^- & a_2^- \\ -a_2^- & a_1^- \end{bmatrix} \qquad \mathbf{b}_- = \begin{bmatrix} b_1^- \\ b_2^- \end{bmatrix}, \qquad \mathbf{c}_-^{\mathrm{T}} = [c_1^- \quad c_2^-], \qquad d^-$$

$$\mathbf{A}_+ = \begin{bmatrix} a_1^+ & a_2^+ \\ -a_2^+ & a_1^+ \end{bmatrix}, \qquad \mathbf{b}_+ = \begin{bmatrix} b_1^+ \\ b_2^+ \end{bmatrix}, \qquad \mathbf{c}_+^{\mathrm{T}} = [c_1^+ \quad c_2^+], \qquad d^+.$$

The elements of $\mathbf{K}_{+-}$ are obtained stepwise from

$$P_R = \tfrac{1}{2}(b_1^+ b_1^- + b_2^+ b_2^-), \qquad P_I = \tfrac{1}{2}(b_1^+ b_2^- - b_2^+ b_1^-),$$

$$Q_R = 1 - (a_1^+ a_1^- + a_2^+ a_2^-), \qquad Q_I = -(a_2^+ a_1^- - a_1^+ a_2^-),$$

$$S_R = \tfrac{1}{2}(b_1^+ b_1^- - b_2^+ b_2^-), \qquad S_I = -\tfrac{1}{2}(b_1^+ b_2^- + b_2^+ b_1^-),$$

$$T_R = 1 - (a_1^+ a_1^- - a_2^+ a_2^-), \qquad T_I = -(a_1^+ a_2^- + a_2^+ a_1^-),$$

$$A_R = \frac{P_R Q_R + P_I Q_I}{Q_R^2 + Q_I^2}, \qquad B_R = \frac{S_R T_R + S_I T_I}{T_R^2 + T_I^2},$$

$$A_I = \frac{P_I Q_R - P_R Q_I}{Q_R^2 + Q_I^2}, \qquad B_I = \frac{S_I T_R - S_R T_I}{T_R^2 + T_I^2},$$

$$k_{11} = A_R + B_R, \qquad k_{12} = A_I - B_I,$$

$$k_{21} = -A_I - B_I, \qquad k_{22} = A_R - B_R.$$

W_{+-} may be computed with the same replacements as above.

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